

Data Cleaning & Preprocessing Pipeline

1. Objective

The goal of this task is to design and document a robust data cleaning and preprocessing pipeline for the dataset conceptualized in previous weeks. The pipeline will simulate cleaning raw data, resolving quality issues, handling missing values, and applying normalization techniques to prepare the dataset for accurate analysis and modeling in the beauty & wellness domain.

2. Data Cleaning Scope

- **Raw Data Simulation:** Assume typical beauty & wellness data (customer profiles, product reviews, sales records).
- **Issues to Address:**
 - Missing values (e.g., incomplete customer details).
 - Outliers (e.g., unrealistic age or purchase amounts).
 - Inconsistencies (e.g., mixed date formats, duplicate entries).
 - Normalization (scaling numerical features for modeling).

3. Step-by-Step Cleaning Process

1. **Data Import:** Load raw data using `pandas.read_csv()` or similar functions.
2. **Missing Values:**
 - Drop rows/columns with excessive nulls (`df.dropna()`).
 - Impute missing values with mean/median/mode (`df.fillna()`).
3. **Outlier Detection:**
 - Use statistical methods (IQR, Z-score).
 - Remove or cap extreme values.
4. **Data Consistency:**
 - Standardize date formats (`pd.to_datetime()`).
 - Normalize categorical labels (e.g., "Male" vs "M").
5. **Normalization:**
 - Apply Min-Max scaling or `StandardScaler` from `scikit-learn`.
6. **Final Dataset:** Save cleaned data (`df.to_csv()`) for reproducibility.

4. Tools & Libraries

- **pandas:** Data manipulation, handling missing values, duplicates.
- **NumPy:** Numerical operations, array handling.
- **scikit-learn:** Preprocessing utilities (scaling, encoding).
- **Matplotlib/Seaborn:** Visualizing outliers and distributions.

5. Pseudo-Code Example

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler

# Load data
df = pd.read_csv("beauty_wellness_raw.csv")

# Handle missing values
df['Age'].fillna(df['Age'].median(), inplace=True)

# Detect outliers using IQR
Q1 = df['PurchaseAmount'].quantile(0.25)
Q3 = df['PurchaseAmount'].quantile(0.75)
IQR = Q3 - Q1
df = df[~((df['PurchaseAmount'] < (Q1 - 1.5 * IQR)) | (df['PurchaseAmount'] > (Q3 + 1.5 * IQR)))]

# Normalize numerical features
scaler = StandardScaler()
df[['PurchaseAmount', 'Age']] = scaler.fit_transform(df[['PurchaseAmount', 'Age']])
```

6. Documentation & Reproducibility

- Maintain a **log of each cleaning step**.
- Save intermediate datasets for traceability.
- Clearly explain rationale behind each decision (e.g., why median imputation was chosen).

7. Flowchart (Text Outline)



8. Impact of Cleaned Data

- Ensures reliable analysis.
- Improves accuracy of predictive models.
- Reduces bias caused by poor data quality.

9. Evaluation Criteria

- Depth & clarity of cleaning process.
- Technical accuracy in Python usage.
- Innovative solutions to domain-specific issues.
- Quality & organization of DOC file.

10. Time Commitment

Estimated **30–35 hours** including planning, coding, documentation, and flowchart creation.